I have uploaded two CSV files:

1. **Generated captions:** Contains image IDs and AI-generated medical captions.
2. **Ground-truth captions:** Contains the corresponding ROCOv2 image IDs and reference captions.

I want you to perform a systematic **Hallucination Assessment** by comparing the generated captions against the ground-truth captions.

I have also uploaded test_concepts.csv, which contains UMLS concept identifiers associated with the ROCOv2 test images.

Use all three files:

- generated_captions.csv
- test_captions.csv
- test_concepts.csv

Match the records using ID.

Use the UMLS concepts as additional reference evidence when assessing whether generated medical concepts are supported or omitted. Recognise medical synonyms, equivalent terminology, negation, laterality and contradictory diagnoses.

Evaluate all matched images and calculate:

- **Unsupported Clinical Concept Rate (%)** = Unsupported generated clinical concepts ÷ Total generated clinical concepts × 100.
- **Clinical Concept Omission Rate (%)** = Omitted reference clinical concepts ÷ Total reference clinical concepts × 100.

Summarising the results as :

- Unsupported Clinical Concept Rate : Assumption Ground Truths Medically Incomplete
- Unsupported Clinical Concept Rate : Assumption Ground Truths Medically Complete
- Clinical Concept Omission Rate : Assumption Ground Truths Medically Incomplete
- Clinical Concept Omission Rate : Assumption Ground Truths Medically Complete
- Confirmed Clinical Contradictions : Across Both Ground Truth Assumptions
- Uncorroborated Clinical Findings : Assumption Ground Truths Medically Incomplete

Provide the results in a simple summary table showing the metric, number of concepts affected, total concepts evaluated and rate (%).

Clearly distinguish confirmed reference mismatches from uncertain cases. Do not count concepts as hallucinations merely because they are absent from incomplete reference descriptions.